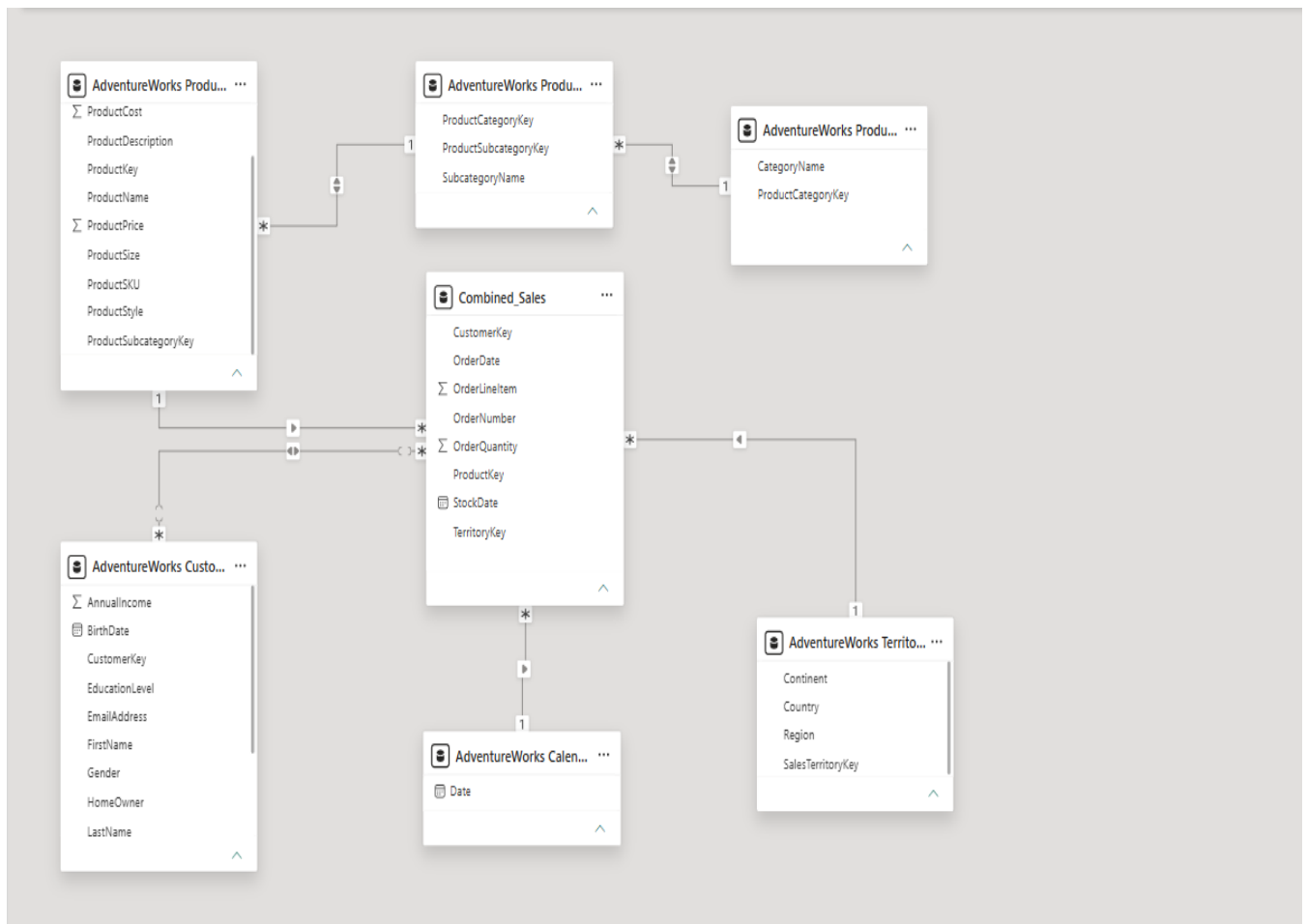


# Data Modeling assignment

Create a star schema by creating relationships between the Sales, Calendar, Customer, Product and Territories tables

Connect all three product tables (Product, Subcategory, Category) in a snowflake schema



Use the matrix visual to confirm that you can filter Order Quantity values using fields from each dimension table.



The image shows a PivotTable in Microsoft Excel. The PivotTable is based on a data source with two fields: 'ModelName' and 'Sum of OrderQuantity'. The PivotTable is filtered to show all items. The data is as follows:

| ModelName               | Sum of OrderQuantity |
|-------------------------|----------------------|
| All-Purpose Bike Stand  | 234                  |
| Bike Wash               | 1706                 |
| Classic Vest            | 521                  |
| Cycling Cap             | 4151                 |
| Fender Set - Mountain   | 3960                 |
| Half-Finger Gloves      | 2644                 |
| Hitch Rack - 4-Bike     | 302                  |
| HL Mountain Tire        | 1305                 |
| HL Road Tire            | 795                  |
| Hydration Pack          | 695                  |
| LL Mountain Tire        | 1560                 |
| LL Road Tire            | 1904                 |
| Long-Sleeve Logo Jersey | 1605                 |
| ML Mountain Tire        | 2119                 |
| ML Road Tire            | 1723                 |
| Mountain Bottle Cage    | 3810                 |
| Mountain Tire Tube      | 5678                 |
| Mountain-100            | 219                  |
| Mountain-200            | 3481                 |
| Mountain-400-W          | 533                  |
| Mountain-500            | 473                  |
| Patch kit               | 5898                 |
| Racing Socks            | 1063                 |
| Road Bottle Cage        | 3329                 |
| Road Tire Tube          | 4327                 |
| <b>Total</b>            | <b>84174</b>         |

#### a) Creating Hierarchies

Create a new hierarchy based on the Start of Year field, and name it “Date Hierarchy” Right-click or drag to add fields until your hierarchy contains the following (in this order):

Start of Year

Start of Month

Start of Week

## Date

Add your new hierarchy to the matrix visual (on rows) and practice drilling up and down between each level of granularity



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| Year         | Sum of OrderQuantity |
|--------------|----------------------|
| 2020         | 2630                 |
| 2021         | 36230                |
| 2022         | 45314                |
| <b>Total</b> | <b>84174</b>         |



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| Month        | Sum of OrderQuantity |
|--------------|----------------------|
| January      | 7446                 |
| February     | 7260                 |
| March        | 7791                 |
| April        | 8174                 |
| May          | 8734                 |
| June         | 8784                 |
| July         | 2201                 |
| August       | 6236                 |
| September    | 6166                 |
| October      | 6610                 |
| November     | 6523                 |
| December     | 8249                 |
| <b>Total</b> | <b>84174</b>         |



| Day          | Sum of OrderQuantity |
|--------------|----------------------|
| 1            | 2778                 |
| 2            | 2832                 |
| 3            | 2736                 |
| 4            | 2820                 |
| 5            | 2813                 |
| 6            | 2762                 |
| 7            | 2684                 |
| 8            | 2844                 |
| 9            | 2819                 |
| 10           | 2731                 |
| 11           | 2780                 |
| 12           | 2755                 |
| 13           | 2711                 |
| <b>Total</b> | <b>84174</b>         |

| OrderDate             | Sum of OrderQuantity |
|-----------------------|----------------------|
| 1/1/2020 12:00:00 AM  | 4                    |
| 1/2/2020 12:00:00 AM  | 4                    |
| 1/3/2020 12:00:00 AM  | 8                    |
| 1/4/2020 12:00:00 AM  | 5                    |
| 1/5/2020 12:00:00 AM  | 3                    |
| 1/6/2020 12:00:00 AM  | 6                    |
| 1/7/2020 12:00:00 AM  | 4                    |
| 1/8/2020 12:00:00 AM  | 8                    |
| 1/9/2020 12:00:00 AM  | 4                    |
| 1/10/2020 12:00:00 AM | 4                    |
| 1/11/2020 12:00:00 AM | 9                    |
| 1/12/2020 12:00:00 AM | 7                    |
| 1/13/2020 12:00:00 AM | 3                    |
| 1/14/2020 12:00:00 AM | 9                    |
| 1/15/2020 12:00:00 AM | 7                    |
| 1/16/2020 12:00:00 AM | 7                    |
| 1/17/2020 12:00:00 AM | 8                    |
| 1/18/2020 12:00:00 AM | 6                    |
| 1/19/2020 12:00:00 AM | 6                    |
| 1/20/2020 12:00:00 AM | 6                    |
| 1/21/2020 12:00:00 AM | 7                    |
| 1/22/2020 12:00:00 AM | 9                    |
| 1/23/2020 12:00:00 AM | 3                    |
| 1/24/2020 12:00:00 AM | 8                    |
| 1/25/2020 12:00:00 AM | 4                    |
| 1/26/2020 12:00:00 AM | 4                    |
| <b>Total</b>          | <b>84174</b>         |